

TASKS:

1. Draw the conceptual model.
2. Identify the possible Entities, their Attributes and the relationships between the entities from the given scenario.
3. Construct your ERD following diagramming convention.
4. Draw the physical model
5. Write the sentences for each relationship, in both directions.
6. Map the relationships
7. Draw the matrix diagram
8. Using Apex, create the table for each entity and insert more than 3 rows in each table

GARDENIT

1. DESCRIPTION

1.1. INTRODUCTION

Over the last few years, the average temperature in Luxembourg has risen in summer and the rate of precipitation has decreased. That's why, nurseries must adapt to completely different situations, as 10 years ago.

You spot a gap in the market in this area and decide to establish a start-up, called GardenIT, to help the nurseries with the new conditions.

Your first potential customer, the large nursery "The Green Thumb", gives you a first order. You should develop a system that allows the gardener to monitor both, the soil moisture and the temperature in outdoor areas or in its greenhouses. Your system should give the gardener detailed information about the soil moisture, to know which areas they would water. Several plants are very sensitive and must therefore not stand to dry or to wet. Otherwise they die off and can no longer be sold.

As the nursery has grown over the last few years and the meteorological conditions have changed, the gardener has difficulty keeping an eye on all their different plants. That is why they asked y to develop a suitable solution for his problem.

2. SCENARIO

2. WEBSITE

Your company's website contains many important points. The customer should be able to log in to the website, adjust his account and read out the data of the modules. Another feature is that your website

should provide to the customer the automatic notification by e-mail for certain situations selected by the customer. (i.e.: If the temperature of a Sensor Station exceeds a certain value).

Your company decides to create the website without a CMS and to develop the complete website itself. The website must be visually appealing and intuitive.

Since customers are often mobile on their properties, it is not enough to adapt the website to computer screens only, but it should also be displayed error-free on mobile phones.

2.1. USER TYPES

To access the actual web page, a user must log in. there are three types of users with the following rights:

1. Company Employees

- A company employee may change his password.
- He still has insight into all sensor data that is assigned to his company.
- An employee can create, edit and delete his or her own notifications.

2. Company Bosses

- As a company boss, you have all the rights that a company employee has.
- In addition, a company boss may create, edit or delete accounts for his employees.

3. Administrators

- Administrators create new companies, edit them and also delete them.
- they are integrating new sensor stations into the system, editing them or deleting them.
- Administrators also have the right to manage all user accounts, i.e. to create new accounts, edit existing ones or also to open accounts.
- Administrators have no right to view the company's data unless your account is tied to a company.

2.2. COMPANIES

A company has the following attributes which can only be changed by an administrator.

- *Name*
- *Address*

2.3. COMPANY MANAGEMENT

Only administrators can access the company administration. Here you can visit:

- create new companies,
- edit companies and
- delete companies.

2.4. USER ACCOUNT

A user account has the following attributes:

- *Email address*: Is set when creating the account, you cannot change or modify it later.
- *Password*: Of course, the password is in encrypted form. The password can be changed or modified later by all user types.

- *Name and first name*: Are set when the account is created and can only be changed or modified by an administrator or company boss.
- *Company*: The company to which the user account is bound. This can only be one or no company at a time. Only administrators are allowed to modify this field.

2.5. USER MANAGEMENT

User management is reserved for administrators and company bosses. The following restrictions must be observed:

1. Administrators can edit all users, companies' bosses only those of their company.
2. Edit means here: create a new account, delete or modify existing ones.
3. You can't delete your own account!
4. A security query must be made prior to deleting an account, because this operation cannot be undone. In addition to the account itself, the user's notification events are deleted.

2.6. SENSORS

Sensors have the following properties:

- *Unique ID*: This is an ID that is integrated into the firmware of the sensor station.
- *Company*: Each sensor station is tied to a company.
- *Name*: In order to be able to identify the individual sensor stations (e.g. according to their location in the nursery), they can be given a name.

2.7. SENSOR-STATION MANAGEMENT

Administrators and company boss have access to the sensor station management. The following restrictions must be observed:

- Company bosses only see their own sensor stations and are only allowed to modify their names.
- Administrators can create new stations, edit and delete existing stations.

2.8. DATA COLLECTION

All data transmitted by the sensor stations are stored in a database, only after checking whether the Sensor Station which as transmitted it, is also registered in the system and after the data itself has been checked for correctness.